

openvaf-r — Full Change Report

Every modification and its reason (original → current)

Ngspice + OpenVAF Enhancements project

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openvaf-r — Full Change Report

Every modification applied to the OpenVAF-Reloaded compiler in this project, with its reason. The baseline is the pristine OpenVAF-Reloaded source tree (as vendored in the project's `original/` snapshot); the current state is the tree committed in this repository under `OpenVAF-master-20260610/`. Each entry links the enhancement write-up in `enhancements_doc/` that carries the full engineering detail and the verifying example suite. The companion document `ngspice_changes_full-report.md` covers the simulator side.

Maintenance note: this report is updated whenever an enhancement touches compiler sources. The per-enhancement index at the end tells you the last change it covers.

Scope summary. ~120 modified source files across the compiler pipeline, two new files (`hir/src/elaborate.rs` — the elaboration pass — and `hir_def/src/item_tree/diagnostics.rs`), plus three classes handled wholesale: the `test_data/` fixtures and snapshots (95 files — new test inputs and snapshots regenerated whenever a feature legitimately changed descriptor contents), the `integration_tests/**/*.osdi` reference objects (regenerated at each ABI bump), and the removal of the AGPL vacask submodule. The compiler builds warning-free; every change below was regression-gated by the example-suite arsenal, the integration suite, and the `VA_TEST` industry-model corpus.

1. Lexer — `openvaf/lexer/` (`lib.rs`, `cursor.rs`)

- Based integer literals (`8'hFF`, `'sb101`): the lexer had only a commented sketch; underscores crashed it. Both the sized and unsized forms now lex, requiring a digit after the base (a silent-0 trap otherwise) ([E-46](#)).
- Don't-care digits `x/X/z/Z/?` accepted in the binary/octal/hex digit runs for `casex/casez` items ([E-78](#)).
- Escaped identifiers (`\my-net!`) lexed per the LRM ([E-46](#)).
- `// comment at EOF` hang fixed: with no trailing newline the line-comment loop spun forever on the EOF character — the only EOF-unsafe loop in the lexer ([E-35](#)).

2. Tokens and grammar — `openvaf/tokens/`, `openvaf/parser/`, `veriloga.ungram`, `sourcegen/`

`tokens/src/parser/generated.rs` gains the keywords and node kinds each language feature needed, threaded through `parser/src/grammar/*` (`stmts.rs`, `expressions.rs`, `items.rs`, `items/module.rs`), the grammar source of truth `veriloga.ungram`, and the `sourcegen/` generators (`ast/src.rs`, `hir_builtins.rs`, `mir_instructions.rs` — the templates the generated token/AST/builtin/instruction tables come from):

- `do ... while` post-test loops ([E-19](#));
- `paramset/endparamset` blocks ([E-21](#));
- `{...}` concatenation and `{n{...}}` replication as real operators, with `'{...}` remaining the typed aggregate ([E-34](#));
- array-literal expressions actually parsed (they were fully scaffolded but unreachable) ([E-4](#), [E-14](#));
- vectored (bus) net declarations and bit-selects ([E-3](#)), multi-index bit-selects for N-D arrays ([E-15](#)), and the LRM name-then-range declaration order ([E-18](#));
- module instantiation items ([E-5](#)), `generate for/genvar` ([E-8](#)), and `generate if/generate case` with optional block labels (previously mis-parsed into a broken `GENERATE_FOR`; [E-67](#));
- `defparam` — a reserved word with no grammar rule ([E-58](#));
- event OR lists `@(cross(...)` or `timer(...))` via a new or keyword and looped event grammar ([E-59](#));
- `casex/casez` sharing the case rule, the keyword token carrying the flavor ([E-78](#));

- `@(initial_step)/@(final_step)` with analysis-phase lists (E-7, E-53);
- indirect branch assignment `V(x): V(y) == expr;` (E-2);
- the `<<</>>>` arithmetic-shift tokens (plus a pre-existing lexer bug fix) (E-6);
- operator precedence corrected against LRM Table 4-2: `%` bound tighter than `*/` (so `6*7%4` evaluated to 18, not 2), and `xnor` split from `xor` (E-38);
- derived-nature parents emitted the wrong node kind (`NAME_REF` where the AST wanted `Path`), leaving the entire inheritance machinery unreachable (E-39).

3. Preprocessor — **openvaf/preprocessor/** (+ **basedb** diagnostics)

- Ten compiler directives that previously hard-failed as undefined macros: ``default_discipline`, ``celldefine/`endcelldefine`, ``unconnected_drive/`nounconnected_drive`, ``timescale`, ``line`, ``pragma`, ``undefineall`, ``default_nettype` (E-6).
- ``default_transition` parsed and threaded to `transition()` lowering (E-47).
- Recursive macro expansion crashed the compiler (stack overflow, direct and mutual): the `MacroRecursion` diagnostic existed but was never emitted, and its report renderer was a `todo!()`. The guard pushes around the macro *body* expansion only — an entry-scoped guard false-positives on the legal `QUAD(x) = TWICE(TWICE(x))` nesting (E-65).

4. Syntax / AST — **openvaf/syntax/** (**ast.rs**, **generated/nodes.rs**, **expr_ext.rs**, **node_ext.rs**, **name.rs**, **validation.rs**, **error.rs**)

- Typed AST nodes for every new construct above (generated from the ungrammar).
- Literal decoding (`expr_ext.rs`): based-int parsing with size/sign handling (E-46) and the mask-aware variant returning `(value, x_mask, z_mask)` for `casex/casez` (E-78); a compiler crash on large bare-integer-shaped literals fixed by falling back to a float constant (E-4).
- String unescaping rewritten as a single pass: the sequential `str::replace` chain corrupted overlapping escapes `\\n` and octal `\\ddd` was missing; all consumers route through one function (E-48).
- **Name::resolve** ate the last character of escaped identifiers (the committed `std.va` snapshot had baked the bug in as `logi`) (E-46).

5. Elaboration — **openvaf/hir/src/elaborate.rs** (new file)

The compile-time flattening pass: every stage downstream of the parser is architected around exactly one flat module per artifact, so hierarchy is resolved by recursively inlining instantiated modules — alpha-renamed per instance, ports bound to the caller's nets, parameters bound to overrides — into an ordinary flat module *before* the rest of the pipeline runs. Built for module instantiation (E-5) and grown into the home of every structural feature:

- `generate for/genvar` unrolling (E-8), nested loops, anonymous blocks, `generate if/case` with elaboration- time constant folding, and the `genvar`-substitution fix (`genvars` in ordinary expressions were routed through identifier renaming, which re-escaped `1e3*(i+1)` into a broken escaped identifier; they now become literal-value holes) (E-67);
- implicit nets from undeclared instance connections, discipline derived from the connected port, prefixed per instance so no cross-instance shorts (E-41);
- hierarchical names `u1.u2.node` and `$root` via an instance-chain map and a hole scanner (E-49);
- **defparam** resolved through the same chain rewrite, with the LRM-mandated precedence over `instance #()` overrides (E-58);
- net concatenation in port connections expanded bit-by-bit onto vectored ports (E-59);
- bus slicing onto instance arrays and bus ports (E-5); `paramset` twin-module rendering (E-21); net initializers preserved through re-rendering (E-45); escaped identifiers re-rendered correctly (E-46).

6. Definition layer — **openvaf/hir_def/** (+ **openvaf/hir/** mirrors)

`body.rs/body/lower.rs/body/pretty.rs`, `expr.rs`, `item_tree*`, `nameres*`, `builtin.rs`, `data.rs`, `types.rs`, `db.rs`, plus the new `item_tree/diagnostics.rs` and the `hir` façade (`body.rs`, `lib.rs`, `db.rs`, `diagnostics.rs`):

- statement/expression collection for every new construct: `do...while` (E-19), `repeat` and `disable` (E-9), `event OR lists` (E-59), `concat/ replication` (E-34), `array aggregates` and `whole-array assignment` (E-14), `CaseKind` + `per-item don't-care masks` with a `stray-literal list` (E-78);
- N-D array machinery: `per-dimension size lists`, `multi-index bit-selects`, `per-element parameter expansion` (E-15); `declaration initializers` on (N-D) arrays via a `Var::array_index` leaf split (E-43);
- **localparam** made genuinely non-overridable (it behaved exactly like `parameter`) with `derived localparams` still tracking inputs (E-9); `paramset lowering` as twin modules whose bound params are `localparams` with `override expressions` (E-21);
- `name resolution`: `electrical ground gnd`; `ordering` (E-9), `nature-attribute access` (`net.potential.abstol` — the third “scaffolded-but-unwired resolver boundary” found) (E-45), `derived-nature resolution` (E-39);
- **builtin.rs** (the system-function registry): the `noise-table pair` (E-9), the full `$random/$dist_*/$rdist_*` family (E-10), `file I/O` and `string functions` (E-11), the last fallback group (`$simprobe`, `aliases`, `plusargs`) — after which `is_unsupported()` is empty (E-12), `$simparam$str` (E-25), `$realtime` (E-59), `$table_model` (E-16);
- new `diagnostics` file: `wrong-arity calls` produced `invalid HIR` and `crashed downstream` — now a proper diagnostic (`functions` and `parameters` both) (E-43);
- multiple `analog blocks` collect into `entry_stmts` in `source order` — the `as-if-concatenated LRM semantics`, by construction (E-60).

7. Types and validation — **openvaf/hir_ty/**

`inference.rs` (+ `inference/fmt_parser.rs`), `builtin.rs` + `builtin/generated.rs` (`signature tables`), `lower.rs`, `types.rs`, `validation.rs`, `validation/body.rs`, `diagnostics.rs`:

- `signature tables` — a recurring defect source: `ANALYSIS` made `variadic` (E-30); `$table_model` given `shape-synthesized varargs signatures` for any dimension (a `generic-varargs fallthrough truncated multi-arity signature lists` — `varargs builtins` with `signature lists` need their own inference arm) (E-40); `TRANSITION's table` was one argument short (3-argument calls crashed, 4-argument worked by accident) (E-47); `transition()` input typed `Real`, a `$dist` typo fixed (E-49); `SIMPARAM_STR` returned `Real` instead of `String` (E-25);
- `format-string inference` (`fmt_parser.rs`): `terminates on every conversion character` and reports it, so `%5d/%-8s/%08x` type their arguments correctly — previously only `real conversions` accepted `flags/width` (E-71);
- `array inference`: `element-wise array case`, `array-literal function arguments` (which previously bound nothing and silently returned 0), `integer arrays` typed `Real`, `output-literal args` accepted silently — all fixed (E-33); `whole-array function arguments/outputs/returns` (E-18, E-20, E-23); `untyped function args` default to `Real` instead of an `ICE` masquerading as an `initializer bug` (E-43);
- **Type::base_type** infinite loop on any `array-type check` (it looped on `self` instead of the `cursor`) — hung the compiler on `simple type mismatches` (E-14);
- `string handling`: `ternary over strings` (`SELECT` + `string binary ops` appended to the `operator tables`) (E-37);
- `validation`: `loop bodies` get their own context so the `analog-operator restriction` reports “loops” (LRM 4.5.1), not “conditions” (E-70); `call-graph cycles` among `analog functions` are `clean errors` naming the cycle instead of a `recursive-inliner stack overflow` (E-59); `casex/casez restrictions` (`stray don't-care literals`, `x digits` under `casez`, `non-integer discriminants`) (E-78); `domain discrete` with `natures` rejected per LRM 3.6.2.2 (E-50); `parameter defaults` exempt from their own

ranges (the CMC "feature disabled" idiom — stock rejected `diode_cmc`, BSIM-CMG, PSP-HV and the HiSIM family at setup) while given values stay fully validated (E-56).

8. Lowering — `openvaf/hir_lower/`

`expr.rs`, `stmt.rs`, `ctx.rs`, `callbacks.rs`, `parameters.rs`, `state.rs`, `fmt.rs`, `lib.rs`:

- analog operators: `laplace_nd/np/zd/zp` via exact state-space realization (E-4) with complex pole/zero pairs per the LRM's (re, im) convention (E-31); `zi_*` via bilinear transform (E-6); `slew/transition` as a saturating tracking loop (E-6) with the LRM negative `max_neg_rate` convention honored (the sign defect made `slew` ignore its input and ramp away) (E-61) and a DC-singularity clamp via an `EnableIntegration` identity (E-47); `idtmod`'s modulo wrap moved off the reactive residual (it diverged) and an argument- index bug fixed (E-27); `idt` initial conditions survive into transient (E-28); the `idt` assert/reset forms with smooth reset dynamics (E-52); `limexp` kept stateless as a documented decision (E-13);
- **`$table_model`**: 1-D piecewise-linear lowered to differentiable MIR so autodiff yields the Jacobian for free (E-16); N-D multilinear as recursive 1-D (E-17, E-40); natural cubic splines with the precomputed moment matrix (E-22);
- events: `cross/above/timer` with persistent previous-value slots (E-8); OR lists folded with a select-based bool-or (a raw `ior` ICEs const-eval) (E-59); final-step gating and phase-list AND-ing (E-53);
- variable persistence: genuine per-instance hidden_state storage behind a two-pass MIR build (E-7), typed slots so integer state stopped crashing instruction selection (E-32);
- `callbacks(callbacks.rs, fmt.rs)`: the `$display` family with the full `[flags][width][.precision]` surface and `%h/%b/%r` translation (E-71); file I/O and string formatting through a generalized `PrintDst` sink (E-11); the deterministic seed-and-salt RNG lowering (E-10); `$simparam$str` (E-25); simulation-control return flags (E-55); `$discontinuity` (E-24);
- operator fixes: unary `~` emitted arithmetic negate; the constant folder sign-extended `>>` where the runtime was unsigned — const and runtime disagreed (E-37);
- masked `casex/casez` comparisons (two `iands` ahead of the existing integer equality) (E-78); array function-argument writeback (E-20) and array returns via a function-named `var-array` (E-23).

9. MIR core and optimization — `openvaf/mir*`

`mir/` (`instructions.rs` + `instructions/generated.rs`, `dfg.rs` with `dfg/phis.rs` and `dfg/uses.rs`, `dominators.rs`, `flowgraph.rs` with `flowgraph/transversal.rs`, `layout.rs`, `cursor.rs`, `lib.rs`, `builder/generated.rs`), `mir_opt/` (`simplify_cfg.rs`, `simplify.rs`, `const_eval.rs`, `global_value_numbering.rs`), `mir_autodiff/` (`builder.rs`, `lib.rs`, `live_derivatives.rs`), `mir_llvm/` (`builder.rs`, `intrinsics.rs`), `mir_build/`, `mir_interpret/`:

- three pre-existing general CFG bugs, all found via event-counter verification: a dangling-reference bug in unreachable-block removal, a multi-exit post-dominance bug in the dominator-tree builder, and a block-merge bug that could corrupt which block the DAE builder treated as the exit (E-8);
- the post-dominator sink-root pathology that let dead-code elimination delete op-dependent `$fatal` calls (side-effecting callbacks under op-dependent control now stay in eval; control dependence computed by arm-reachability) (E-55);
- const-eval agreement with runtime semantics for shifts and the bitwise-not fix (E-37);
- autodiff: noise operators process before any ddt (a shared ddt evaluated between two noise operators silently dropped the second source), and the reactive coupling of a ddt-shaped noise wave is registered so small-signal pruning keeps it (E-54);
- **`llvm.fabs.f64`** registered in the LLVM intrinsics table — it never had been, and the signed noise-power convention needed it (E-42);
- new opcodes/instructions supporting the operator work (barrier/ integration-identity forms; MIR deliberately has no fabs, so lowering composes it) (E-47, E-52).

10. DAE construction — **openvaf/sim_back/**

dae.rs + dae/builder.rs, topology* (incl. linalize.rs, small_signal_network.rs), noise.rs, context.rs, lib.rs:

- implicit equations for indirect branch assignment (one new unknown – residual per statement) (E-2) and the absdelay synthetic two-node form (E-1) — with equation indices kept stable by mapping dead/collapsed unknowns to zero-contribution placeholders instead of renumbering;
- port-flow probes **I(<port>)**: a TODO stub returning 0 became a synthesized DAE unknown mirroring the node's KCL residual (E-29);
- probe-only branches: probing a never-contributed branch read 0 and an open circuit — a 0 V-source pass materializes them, enabling ideal ammeters and flow-only signal-flow disciplines (E-36);
- noise factors (noise.rs): equation-path noise was silently lost (never attached to the DAE); factors generalize to $re + j\omega \cdot im$ with no extra matrix unknowns (E-54); same-named source grouping metadata (E-42);
- idt reset-mode charge dynamics and conditional step bounding (E-52).

11. OSDI code generation — **openvaf/osdi/**

lib.rs, metadata.rs + metadata/osdi_0_4.rs, inst_data.rs, eval.rs, load.rs, setup.rs, compilation_unit.rs, ndatable.rs, stdlib.c, build.rs, reference headers:

- descriptor emission for every ABI evolution (see the ngspice report's ABI table): the absdelay/last_crossing info arrays (E-1, E-6), `OsdiNode.nodeset` (E-45), the `ac_stim` source array + `load_ac_stim` (E-51), stride-2 signed noise pairs in `load_noise` (E-54) — the version tuple lives in `osdi/src/lib.rs`;
- **eval.rs**: final-step flag gating (E-53); simulation-control return flags (E-55); the `ac_stim` large-signal value (was an unreachable!() panic) (E-26);
- **inst_data.rs**: absdelay slot layout (E-1); hidden-state slots typed by the variable (`ty_f64` was hardcoded — integer state fed f64 loads into integer MIR ops and aborted instruction selection) (E-32);
- **compilation_unit.rs** (print codegen): the `%b` segfault — the binary-formatted string was remembered for `free()` but never pushed to the `snprintf` argument list, so the matching `%s` consumed a garbage pointer (E-71); the print-callback machinery with its shadowed-fun and volatile IPO traps (E-11);
- **stdlib.c**: the file-descriptor table behind `$fopen/$fdisplay/...` (E-11); the `splitmix64` `osdi_rng_*` runtime (E-10); a `simplparam_str` loop/return bug (E-25);
- **ndatable.rs**: ddt/idt natures resolved through `resolve_nature_index` instead of panicking on derived natures (E-39);
- **setup.rs**: range validation with default-exemption (E-56); a 64-line abandoned unsafe experiment (`VoidAbortCallback`) deleted in the warning cleanup (E-66);
- **build.rs**: the copied-target `stdlib` compilation trap fixed (E-10).

12. Driver, libraries, and infrastructure

- `openvaf-driver/src/crash_report.rs`, `linker/src/lib.rs`, `lib/bforest/` (`map.rs`, `pool.rs`, `set.rs`), `lib/typed_indexmap/` (`set.rs`): the zero-warning cleanup (`44 → 0` across 13 crates: lifetime annotations, dead code, nightly cfgs, an unused LLVM-prefix probe) (E-66);
- `basedb/` (`ast_id_map.rs`, `lints.rs`, `lib.rs`, `diagnostics/preprocessor_error.rs`, `diagnostics/syntax_error.rs`): AST-id plumbing for elaboration-created items (the `AstId::from_erased` placeholder used by array returns; E-23), preprocessor/syntax diagnostic rendering for the new errors;

- `.llvm-version` pins the LLVM 18 toolchain expectation; the `hir` and preprocessor `Cargo.tomls` carry the dependency additions their new code needed.

13. The test suite — `openvaf/openvaf/tests/`, `lib/mini_harness/`, `test_data/`, `integration_tests/`

The fork's own integration suite over real compact models had never run in this project: its OSDI loader was frozen at ABI 0.4, its optional VACASK legs panicked on a never-initialized submodule, and the tests hide behind `RUN_DEV_TESTS=1`. Fixed test-side only (E-68):

- `tests/load/osdi_0_4.rs` + `load/mod.rs`: loader structs synced to ABI 0.7; `mock_sim/mod.rs`: stride-2 noise convention; `mini_harness`: missing directories skip with a note instead of panicking;
- VACASK removed entirely (`.gitmodules`, the submodule, 34 harness legs and their stale snapshots): AGPL-3.0 cannot be vendored here;
- `test_data/` (95 files as a class): new fixtures for new features and snapshots regenerated whenever descriptor contents legitimately changed — every regeneration reviewed (the diffs are the project's own features: `flow(<port>)` unknowns, probe-only branches, implicit-equation nodes); `integration_tests/*/*.osdi` regenerated at each ABI bump.

14. Per-enhancement index (compiler-touching enhancements)

Enhancement	Pipeline areas	One line
E-1	<code>sim_back</code> , <code>osdi</code>	<code>absdelay()</code> synthetic-node DAE + descriptor arrays
E-2	<code>parser</code> → <code>hir_lower</code> , <code>sim_back</code>	indirect branch assignment (implicit equations)
E-3	<code>parser</code> , <code>hir_def</code>	vectored (bus) nets with bit-selects
E-4	<code>hir_lower</code> , <code>syntax</code> , <code>parser</code>	<code>laplace_*</code> state-space + array-literal parsing
E-5	<code>elaborate.rs</code> (new), <code>parser</code>	module instantiation via compile-time flattening
E-6	<code>preprocessor</code> , <code>parser</code> , <code>hir_lower</code> , <code>osdi</code>	directives, shifts, slew/transition, <code>zi_*</code> , <code>last_crossing</code>
E-7	<code>hir_lower</code> (state), <code>osdi</code>	<code>@(initial_step)</code> gating + variable persistence
E-8	<code>parser</code> , <code>elaborate</code> , <code>hir_lower</code> , <code>mir/mir_opt</code>	generate for + events; three general CFG bug fixes
E-9	<code>hir_def</code> , <code>hir_lower</code> , <code>sim_back</code>	noise tables; <code>localparam/ground/strings</code> ; <code>repeat/disable</code>
E-10	<code>hir_lower</code> , <code>osdi</code> <code>stdlib</code>	<code>\$random/\$dist_*</code> deterministic RNG
E-11	<code>hir_lower</code> , <code>osdi</code>	file I/O + string functions (<code>PrintDst</code>)
E-12	<code>hir_def</code> , <code>hir_lower</code>	last unsupported builtins as LRM fallbacks
E-13	<code>hir_lower</code>	<code>limexp</code> kept stateless (documented decision)
E-14	<code>hir_def</code> , <code>hir_ty</code> , <code>hir_lower</code>	array literals/params/dynamic indexing; <code>base_type</code> hang
E-15	<code>hir_def</code> , <code>hir_ty</code> , <code>hir_lower</code>	N-D arrays
E-16/17/22/40	<code>hir_ty</code> , <code>hir_lower</code>	<code>\$table_model</code> : 1-D, N-D multilinear, cubic spline, <code>varargs</code> sigs
E-18/20/23/33	<code>hir_ty</code> , <code>hir_lower</code> , <code>basedb</code>	arrays in analog functions: <code>in/out/return/literals</code> + array case
E-19	<code>tokens</code> → <code>hir_lower</code>	<code>do ... while</code>
E-21/44	<code>parser</code> , <code>elaborate</code> , <code>hir_def</code> , <code>sim_back</code>	<code>paramset</code> + hidden system parameters

Enhancement	Pipeline areas	One line
E-24/55	hir_lower, mir, osdi	\$discontinuity; \$finish/\$stop/\$fatal honored
E-25	hir_ty, osdi stdlib	\$simparam\$str
E-26/51	osdi, sim_back	ac_stim: crash fix, then full AC-RHS injection
E-27/28/52	hir_lower, sim_back	the idt family: modulo, IC, assert/reset
E-29/36	sim_back	port-flow probes; probe-only branches
E-30	hir_ty, hir_lower	variadic analysis()
E-31	hir_lower	complex poles/zeros in root forms
E-32	osdi inst_data	integer persistent state (crash fix)
E-34	tokens→hir_lower	{...} concat + {n{...}} replication
E-35	lexer	EOF comment hang
E-37/38	hir_lower, mir_opt, parser	operator + precedence audits (~, >> fold, % level)
E-39	parser, hir_ty, osdi	derived natures unwired at the node-kind boundary
E-41/49/58/59	elaborate	implicit nets; hierarchical names; defparam; port concat + OR lists
E-42/54	sim_back noise, mir_autodiff, mir_llvm, osdi	correlated + node-free complex noise factors
E-43	hir_def (+diagnostics), hir_ty	initializers completed; arity diagnostics
E-45	osdi, hir_def, elaborate	nodesets + nature-attribute access (ABI 0.5)
E-46/48	lexer, syntax	escaped ids, based literals, single-pass unescaper
E-47	preprocessor, hir_ty, hir_lower	`default_transition + transition fixes
E-50	hir_ty validation	domain-binding validation
E-53	hir_lower, osdi eval	@(final_step) + phase lists
E-56	hir_ty, osdi setup	parameter defaults exempt from ranges
E-59	tokens→hir_lower, hir_ty	\$realtime; OR lists; recursion diagnostics
E-61	hir_lower	slew sign fix (operator-args audit)
E-65	preprocessor	macro-recursion guard
E-66	13 crates	zero-warning cleanup (44 → 0)
E-67	tokens, parser, syntax, elaborate	generate audit: genvar fix, nesting, if/case
E-68	tests, mini_harness	integration suite enabled; VACASK removed
E-70	hir_ty validation	loop-context diagnostics
E-71	hir_ty fmt, hir_lower fmt, osdi	display-format surface + %b segfault
E-78	lexer→hir_lower, hir_ty	casex/casez don't-care masks

Enhancements not listed (57, 60, 62–64, 69, 72–77, 79–81) changed no compiler sources — they were validation suites, documentation, benchmark tooling, or ngspice-side work (see the [ngspice report](#)).